

An online fuzzy fraud detection framework for credit card transactions

Georgios Charizanos^a, Haydar Demirhan^{a,*}, Duygu İçen^b

^a RMIT University, Mathematical Sciences, School of Science, Melbourne, 3000, Australia

^b Hacettepe University, Department of Statistics, Ankara, 06800, Türkiye

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ABSTRACT

Credit card transaction fraud is one of the challenging security concerns for financial firms globally. The continuously changing environment of fraud characteristics and the class imbalance and complete separation issues in fraud data create difficulties in accurately and efficiently predicting fraudulent transactions and implementing fraud detection systems in real-time. The study aims to develop a new real-time fraud detection framework that can efficiently be implemented online and address the issues caused by non-stationary changes in transaction and fraud characteristics, class imbalance, and complete separation. We propose a new approach to handle the impact of non-stationary changes in fraud transaction patterns. It enables efficiency in model training, given the sheer size of datasets. By implementing a robust fuzzy logistic regression model against class imbalance and separation problems, we address the challenge of having a very low rate of fraudulent transactions in the dataset and having separation issues due to specific characteristics of transactions. The analysis of the performance versus efficiency nexus of the proposed methodology reveals that the proposed framework shows strong performance results with specificity and sensitivity greater than 0.90 and Matthew's correlation coefficient greater than 0.80 even on small sample sizes and produces highly accurate results in identifying fraudulent and non-fraudulent transactions with an accuracy greater than 0.99. Benchmarking with machine learning and other fraud detection approaches reveals that the proposed framework provides better detection performance while maintaining a higher rate of identifying non-fraudulent transactions than the alternative approaches. Improved classification performance leads to detecting fraudulent transactions with higher precision while avoiding misclassifying legitimate transfers, resulting in lower financial losses and improved customer satisfaction.

1. Introduction

Credit card fraud poses a significant financial burden on the global economy. 3.5 billion fraudulent card transactions were recorded in 2022 globally, up from just 1.5 billion in 1992, which resulted in a total loss of 321 billion US dollars (Aite-Novarica, 2022; Nilson-Report, 2024). Furthermore, the frequency of fraud activity has grown by 133% in the last 30 years. For example, in Australia, 1.7 million people were victims of fraud activity in 2021 and 2022, which is equivalent to 8.1% of the country's total population (ABS, 2023). Credit card fraud involves the unauthorized use of cards linked to legitimate consumer accounts. It can be detected as unauthorized transactions on a customer's banking accounts, where a transaction is usually flagged as fraudulent ($Y = 1$) or not ($Y = 0$). In response to the escalation of fraudulent activities, it is imperative to harness capabilities from data science methodologies to facilitate precise predictions of such events.

There are three key characteristics of credit card fraud detection in the literature: Firstly, fraud characteristics are non-stationary and

constantly change over time. The detection systems aim to identify anomalies in transaction data composed of different characteristics of fraudulent activity (Dal Pozzolo, Caelen, Le Borgne, Waterschoot, & Bontempi, 2014). These datasets can either be historical or real-time live transaction data. The dynamic nature of fraud characteristics is due to the changing behavior of fraudsters, different types of fraud, and consumer demographics that influence human behavior towards fraud (Kurshan, Shen, & Yu, 2020). This dynamic nature can cause fraud characteristics to change rapidly at any point in time, leading to the need for models that can handle rapid changes (Kitts et al., 2015). Secondly, these systems utilize analytical methods to identify fraudulent and non-fraudulent transaction patterns in the binary response variable. However, such methods are sensitive to the problems of class imbalance and complete separation, which can result in biased predictions. Finally, such systems need to operate on a real-time basis. Over 25 million transactions occur each hour globally (Sandberg, 2023). Fraud detection algorithms must efficiently and accurately operate on

* Corresponding author.

E-mail addresses: S3832990@student.rmit.edu.au (G. Charizanos), haydar.demirhan@rmit.edu.au (H. Demirhan), duyguicn@hacettepe.edu.tr (D. İçen).

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large-scale data to achieve real-time implementation. The immense size of transaction data is a significant obstacle to the efficiency of implementation and, hence, the security of consumer accounts, given that it is often challenging to protect consumers retrospectively.

The core methods to detect fraudulent anomalies in transaction data include machine learning (ML), statistical, and fuzzy approaches. A systematic review of literature on credit card fraud detection using approaches such as support vector machines (SVMs), random forests, neural networks (NNs), deep learning methods, K-nearest neighbors (KNN), and extended gradient boosted tree is presented by [Ghai and Kang \(2022\)](#). Another extensive systematic literature survey by [Abdallah, Maarof, and Zainal \(2016\)](#) provides a comprehensive overview of issues impacting the performance of fraud detection systems.

[Hilal, Gadsden, and Yawney \(2022\)](#) explore SVMs in detail, proposing seven variations ranging from classical SVMs to incremental clustering, weighted one-class discriminative feature selection, and KNN models. A classification performance analysis is given by assessing the overall accuracy of each method applied to machine learning datasets. [Shen, Tong, and Deng \(2007\)](#) focus on using decision trees against NNs and classical logistic regression on a credit card transactions dataset, which is not freely available, to compare the performance of considered methods. [Fang, Zhang, and Huang \(2019\)](#) propose over-sampling in ML methods to improve classification performance given the imbalance problem in transaction fraud data. They compare a light gradient boosting model against random forests and gradient boosting machines, indicating that the former produces stronger classification results with better efficiency. [Strelcenia and Prakoonwit \(2023\)](#) explore various data augmentation methods, such as kernelized conditional generative adversarial networks. The authors focus on how such methods perform under the significant imbalance setting for fraud detection, showcasing strong F1 and accuracy scores. Similarly, [Ba \(2019\)](#) uses generative adversarial networks (GANs) for fraud detection using real credit card transaction data from 2015, which is also used in this study. [Chen, Lee, Huang, and Peng \(2024\)](#) propose using intelligent sampling with a self-supervised learning algorithm for the binary classification of two problems involving credit card fraud and card default rates. The authors explore the use of such machine learning techniques against imbalance problems and real-time implementation.

The classical statistical approach in fraud detection involves using logistic regression for the classification task ([Ghai & Kang, 2022](#); [Shen et al., 2007](#)). However, classical logistic regression showcases a low fraud detection rate with significant inaccuracy, which can be attributed to the fact that classical logistic regression methods are vulnerable to imbalance and separation issues, resulting in low classification accuracy ([Charizanos, Demirhan, & İcen, 2023](#)).

Fuzzy methods constitute another class of methods employed for detecting fraudulent transactions. [Lenard and Alam \(2005\)](#) analyze using fuzzy logic and fuzzy clustering methods for fraud detection. The authors explore both the benefits and critical challenges of fuzzy methods for fraud detection from a theoretical perspective, concluding that fuzzy methods can offer significant opportunities for fraud detection for online card transactions. [Casalino, Castellano, and Mencar \(2019\)](#) consider using fuzzy techniques for fraud detection, such as dynamic incremental semi-supervised fuzzy c-means clustering. The application is done on the same credit card transaction dataset as in this study. The authors conclude that the proposed fuzzy method performs well in handling the imbalance problem. [Behera and Panigrahi \(2015\)](#) propose using a fuzzy NN model that involves an input fuzzy rule combiner component with an NN learning component, which shows promising results in handling imbalance and classification accuracy. We compare the model proposed in our study against [Behera and Panigrahi \(2015\)](#)'s work. However, it is not possible to compare the performance of our model to [Casalino et al. \(2019\)](#)'s model since they do not cover the performance measures we consider. [Razooqi, Khurana, Raahemifar, and Abhari \(2016\)](#) propose a fuzzy logic methodology for fraud detection, which is compared against artificial neural networks

(ANNs). The authors analyze the fuzzification and de-fuzzification of input and output information used to predict probabilities in a fuzzy logic model, then implement this model in five steps. The classification results, evaluated using only mean squared error, indicate that the ANN model outperforms the fuzzy method. However, multiple measures are required to provide targeted insights towards a confusion matrix. [Jain, Tiwari, Dubey, and Jain \(2019\)](#) explore several algorithms used for fraud detection, including SVMs, ANNs, KNN, decision trees, and a fuzzy logic methodology. The authors showcase a confusion matrix approach to validate classification performance. The ANN model outperforms all other considered methods using a percentage accuracy score. However, using a single performance measure does not ensure generalizability and only provides insights into the model's overall accuracy. It does not provide a complete assessment of how these models perform under imbalance and separation.

Hybrid models are also considered for fraud detection in the literature. [Dal Pozzolo, Boracchi, Caelen, Alippi, and Bontempi \(2017\)](#) and [Dal Pozzolo et al. \(2014\)](#) use a two-step method with two semi-supervised models. One model is implemented online in a smaller transaction sample size, and the other is used to train the model on the whole dataset. This involves implementing the method in real-time using a small sample to train the model, followed by a retrospective analysis on a much larger dataset to validate model parameter selection and performance. [Srivastava, Kundu, Sural, and Majumdar \(2008\)](#) consider using hidden Markov models (HMMs), trained by using behavioral statistics of cardholders in transaction data. The authors use a process of probabilistic elimination with the HMM model, where if a transaction does not produce a sufficiently high probability, it is classified as fraudulent. [Maes, Tuyls, Vanschoenwinkel, and Manderick \(2002\)](#) propose using a Bayesian belief network model instead while they compare against an NN approach. However, the Bayesian and NN models showcase a low true positive rate, indicating a weak fraud detection accuracy. [HaratiNik, Akrami, Khadivi, and Shajari \(2012\)](#) propose a hybrid model that utilizes Fogg behavioral analysis on the merchants, assessing their motivation and ability towards card fraud. Information from this analysis is then used to obtain fraud tendency weights, which predict fraudulent activity, given a set of fuzzy rules. The model showcases strong true positive rates against standard fuzzy methods with a moderately high false positive rate. While this could be attributed to imbalance issues, the authors do not use a comprehensive set of performance measures to explore the impact of imbalance and separation.

Critical limitations to these analytical methods encompass several important aspects. There is a lack of focus on imbalance issues in fraud detection studies ([Behera & Panigrahi, 2015](#); [Ghai & Kang, 2022](#); [Maes et al., 2002](#)). Transaction data are significantly imbalanced towards non-fraudulent activities. The presence of a class imbalance in the data can bias the model to predict mostly the majority class. This leads to poor performance in detecting fraudulent activity, which is the minority class in the binary responses. It may also lead to a model that is less sensitive to detecting the true underlying patterns and associations in fraudulent transactions, hence more vulnerability towards false negative predictions. While a few studies for fraud detection mention the imbalance issue ([Abdallah et al., 2016](#); [Ba, 2019](#); [Hilal et al., 2022](#)), these do not explore the details of any potential impact. [Behera and Panigrahi \(2015\)](#) evaluate a hybrid of a rule-based fuzzy inference system and NN for fraud detection by utilizing Receiver Operating Characteristic (ROC) curves to assess classification performance. However, due to the lack of consideration of the specificity and sensitivity, it is unclear how the imbalance issues in the data influence the proposed model's classification performance. [Ghai and Kang \(2022\)](#) explore several different ML methods to propose solutions to the fraud detection problem. However, there is no consideration of the class imbalance problem in [Ghai and Kang \(2022\)](#)'s study, which could be achieved by employing a comprehensive assessment of the sensitivity and specificity performance measures. [Hilal et al. \(2022\)](#)

provide a thorough discussion of the class imbalance issue in fraud detection studies, exploring how the proposed SVM algorithms provide a robust solution to the imbalance problem. The authors comment that this occurs due to the ability of such methods to work with many different data features. However, the lack of focus on the sensitivity and specificity performance of the proposed classification methods showcases a lack of extensive analysis on how imbalance actually impacts classification performance on these SVMs. [Abdallah et al. \(2016\)](#) consider several different ML models in a survey of methods for fraud detection. The authors explore several different algorithms and the core problems associated with online fraud detection, specifically defining the imbalance setting to be generally different based on the type of fraudulent activity. However, the class imbalance problem needs to be considered from a classification performance perspective. On the other hand, [Strelcenia and Prakoonwit \(2023\)](#) thoroughly discuss how class imbalance impacts classification performance and claim that the proposed GAN methodologies are robust towards this issue. However, there is no analysis of the sensitivity and specificity performance of the proposed algorithm, which would enable a complete assessment of how well the model performs under the imbalance setting. Similarly, [Ba \(2019\)](#) offers limited consideration of the imbalance problem with no direct analysis of how the GAN algorithm performs against the significant imbalance towards fraudulent activity. The authors showcase an extensive analysis of classification performance, providing greater insights into how the model performs in general. However, the exclusion of sensitivity and specificity measures does not provide an additional layer of assessment targeted at the imbalance problem. Moreover, the class imbalance problem is often approached from a theoretical standpoint with only accuracy and F1 scores being considered ([Strelcenia & Prakoonwit, 2023](#)). However, in summary, this crucial issue needs to be assessed and discussed thoroughly as part of analyzing the model results using a comprehensive set of performance measures that include both sensitivity and specificity scores.

The lack of consideration on the issue of complete separation and its impact on classification performance is another limitation in the literature. Separation can occur even on significantly imbalanced datasets. For example, consider fraudulent activity that could be associated with higher transaction amounts, specific geographic locations, or specific consumer demographics, such as age or profession. This could mean that certain consumer populations are more vulnerable to online fraud; hence, separation may occur. Even if fraudulent activity is the minority class with significant imbalance, these characteristics would perfectly separate the probability of a fraudulent transaction. The presence of complete separation leads to a model that fits the training data exceptionally well but performs poorly with data outside the training set due to being unable to learn from meaningful patterns in the data. Therefore, separation should be considered while assessing classification performance. However, while [Abdallah et al. \(2016\)](#) and [Hilal et al. \(2022\)](#) explore the problem of class imbalance to a great extent from a theoretical perspective, there is no mention of the problem of complete separation in fraud datasets and how this could influence classification performance through over-fitting issues. [Ba \(2019\)](#) claims that the proposed methodology is robust towards imbalance problems, with strong performance scores across several measures. However, there is no exploration of how classification performance could be artificially inflated due to over-fitting caused by separation issues. Lastly, while [Strelcenia and Prakoonwit \(2023\)](#) provide an extensive discussion on the imbalance issue, the authors do not explore how the presence of separation would impact classification performance. Moreover, several instances could benefit from analyzing the separation issue and the concept of over-fitting ([Behera & Panigrahi, 2015](#); [Shen et al., 2007](#)).

The reviewed literature showcases a limited number of measures used to assess the overall performance of the implemented models. [Behera and Panigrahi \(2015\)](#) only utilize the ROC curves to assess classification performance, while [Shen et al. \(2007\)](#) consider a cumulative rate

of decline percentage score. On the other hand, [Strelcenia and Prakoonwit \(2023\)](#) consider an extensive selection of classification performance measures, such as precision, recall, F1 score and accuracy. However, the lack of specificity and sensitivity scores do not allow for a complete assessment of imbalance, while Mathew's Correlation Coefficient (MCC) would have provided further insights towards separation. While [Chen et al. \(2024\)](#)'s approach showcases strong F1 performance across both datasets, the set of performance measures is not sufficient to fully assess the response of proposed algorithms to the imbalance and separation problems. The authors provide only F1, precision, and recall as measures, which do not fully showcase that the model responds well to the imbalance problem. This is due to the lack of analysis on variability between performance sensitivity and specificity, which would be a strong indicator of handling imbalance problems. Moreover, the lack of sufficient performance measures does not allow for a comprehensive assessment of potential over-fitting caused by complete separation in the data.

A comprehensive set of measures such as sensitivity, specificity, MCC, and F1 score provide a more thorough understanding of the combined influence of class imbalance with complete separation on a classifier. The presence of perfect scores across specificity, sensitivity, and F1 scores strongly indicates that a model learns extensively from a distinct group of predictors, hence not identifying other patterns successfully. The limited number of performance measures employed per study in the literature indicates the need for such an assessment. Moreover, the small set of performance metrics and lack of sensitivity and specificity assessment make it difficult to provide targeted insights towards imbalance issues ([Charizanos et al., 2023](#)). The studies that do not consider sensitivity and specificity together potentially overlook the impact of class imbalance and report a model as promising while it is severely impacted by separation.

Moreover, there is limited consideration of the non-stationary changes in fraud characteristics. Fraud transaction data include several consumer and merchant characteristics that influence fraudster and consumer behavior towards credit card transactions. These rapid changes in behaviors impact how efficiently and accurately a financial institution can react to fraudulent activity. As such, these characteristics of fraudulent activity can also influence the classification performance of any given model when applied in real-time. [Kursan et al. \(2020\)](#) focus on the non-stationary characteristics of fraud through the lenses of the different types of fraudulent activity and how these influence behavior. They identify and explore the eight most common types of fraud, ranging from online payments to fraudulent transactions due to identity theft and elderly scams. It is shown that different fraud types impact consumers differently based on certain demographic elements. Furthermore, the authors explain that such non-stationarity creates rapid changes in fraud patterns and conclude that such issues cause significant scaling and complexity challenges in fraud data due to the variability in fraud characteristics. [Dal Pozzolo et al. \(2017\)](#) explore different characteristics of fraudulent transactions, including consumer and merchant information. The authors define these as demographic instances of consumer and merchant and financial nature that describe a transaction. These instances are non-stationary, given the vast variability of characteristics that cause fraud patterns to occur rapidly, even in short periods. [He, Leslie, and Grant \(2024\)](#) explore the nature of non-stationary characteristics in fraudulent activity on a time series basis. The authors propose a hybrid approach with a classical time series and NN methodology applied alongside a fuzzy clustering model. The non-stationary characteristics of fraud are only defined in this study through the concept of a time series format. At the same time, the authors do not consider how the different types of fraud can influence rapidly changing patterns.

Lastly, there is a lack of exploration on how the proposed models can be implemented on a real-time basis. Some works do not consider how the proposed methods can be implemented in real-time ([Srivastava et al., 2008](#); [Strelcenia & Prakoonwit, 2023](#)). On the other hand,

one proposition involves a two-step method with two semi-supervised models (Dal Pozzolo et al., 2017, 2014). The first step includes running the method in real-time on a small sample of transactions, and the second step is to run the model retrospectively on the full dataset to adjust the first. However, such an approach may be susceptible to non-stationary changes in fraud characteristics because it relies heavily on learning from past data in equal terms with recent data. Another methodology involves using ML methods such as random forest, SVMs, and decision trees on historical data to train the model in recognizing cardholder, transaction, and merchant features. These features then form patterns to predict whether a transaction is fraudulent or not on a real-time basis. However, the overall complexity of the model and consideration of limited fraud characteristics do not allow for adaptability on real-time detection, given the continuously changing patterns of fraud in the finance industry. Moreover, deep learning techniques are used to detect patterns in real-time (Abakarim, Lahby, & Attioui, 2018). An auto-encoder, an NN trained to reconstruct input data and learn key patterns in non-fraudulent transactions, is used by Abakarim et al. (2018). However, while the model is trained on historical fraud data and then deployed for real-time classification, the computational effort requirement is unclear. As the classification of live transactions occurs in one main stage, there can be challenges with computation time if large datasets are processed. The overall method of historical data retrieval using structured query language pipelines can increase complexity, with such systems being sensitive to changes in the underlying data structures. These challenges can result in inefficiencies, hence, the inability to detect fraud in real-time.

The challenges associated with the mentioned issues can detrimentally impact predictive accuracy, causing significant losses due to fraudulent transactions or inconvenience to customers who are making legitimate transactions classified as false positives. One method for reducing false positive rates in fraud detection systems utilizes tree-based rule generation ML algorithms, such as random forest, decision tree, and gradient boosting (Vorobyev & Krivitskaya, 2022). While such approaches can help reduce false positives, they do not address the issues of false negative rates. This inflates fraud risk due to incorrectly classified fraudulent transactions. Nevertheless, these limitations highlight the need for further research into implementing models in real-time, assessing models appropriately for separation, imbalance, and overall comprehensive performance and optimization evaluation.

The primary motivation of our work is to propose a new methodology for real-time fraud detection that specifically addresses the critical challenges identified in the literature. The focus is on the imbalance, complete separation, and efficiency for real-time implementation issues. Meanwhile, we address the rapid changes in fraud characteristics. We utilize the fuzzy logistic regression framework of Charizanos et al. (2023), which is shown to be robust against imbalance and separation issues. We implement this approach in three stages: tuning, learning, and online implementation. This framework offers essential benefits in handling class imbalance and complete separation in the data, enhanced interpretability through fuzzy membership values, flexibility in decision boundaries, and applicability of the fuzzy logistic regression framework on a real-time environment with large data (Charizanos et al., 2023; Pourahmad, Ayatollahi, Taheri, & Agahi, 2011). Thus, this study's central objective is to develop a real-time transaction fraud detection methodology that is not impacted by class imbalance and separation problems, provides practitioners with low false positive and high true positive detection rates, captures changing, non-stationary fraud characteristics in the data, and can operate on smaller sample sizes without losing performance. This is achieved by using the fuzzy classifier from Charizanos et al. (2023) and separately employing tuning and learning stages using weighted historical data and then implementing online fraud detection on the most recent transaction data.

The practical contributions of our study are that (i) it provides a highly precise and efficient solution for transaction fraud detection

using the fuzzy logistic regression framework on a real-time basis. (ii) The proposed method handles the imbalance in the proportions of fraudulent and legitimate transactions in the data without losing accuracy. (iii) It is unaffected by the separation created due to data characteristics. (iv) Our method successfully captures the changing fraudster behavior and consumer characteristics to provide consistently high precision. (v) Our framework is readily applicable in banking practice as all the computer codes to implement the framework are available at <https://github.com/GZanos/FLL>. From the methodological viewpoint, the contributions of our study are that (i) we utilize a probabilistic sampling technique to create the model's training dataset that allows us to capture changing fraud characteristics better and to reduce the size of the required training dataset to increase the model's efficiency in online implementation. (ii) The use of fuzzy coefficients and the fuzzification of the binary responses allow the model to account for more uncertainty, resulting in better handling of class imbalance and separation issues in fraud data.

Section 2 describes the data and proposed methodology in this study. Section 3 depicts the application of the proposed online fraud detection framework and compares it with other fraud detection approaches that use the same fraud dataset. Section 4 includes the study's conclusions and discussions about its limitations and future work.

2. Data and methodology

2.1. Real transaction fraud dataset

We utilize a European customer transaction dataset, which is a real dataset of credit card transactions, including 284,807 transactions done over two days, 492 of which are fraudulent. This real dataset is also used or analyzed in many studies (Carcillo, Dal Pozzolo et al., 2018; Carcillo, Le Borgne, Caelen and Bontempi, 2018; Carcillo et al., 2021; Dal Pozzolo et al., 2017, 2014; Han, Zhu, Zhou, & Cai, 2022; Lebichot, Le Borgne, He-Guelton, Oblé, & Bontempi, 2020). The dataset contains 29 predictors, including the amount of each transaction, named V29. The total value of the transactions in the data is estimated to be \$25.2 million US dollars, with the average transaction amount varying from \$0.01 USD to \$25,692 USD. The average amount for non-fraudulent transactions is \$88 USD, while for fraudulent transactions, it increases to \$122 USD. The dataset has 28 more characteristics, named from V1 to V28, that are masked due to data sensitivity policy in financial institutions.

As expected, the number of fraudulent transactions relative to the size of the data indicates that the data is significantly imbalanced towards normal transactions ($Y = 0$ class) in the binary response variable, given that only 0.17% of the transactions are actually fraudulent ($Y = 1$ class).

We utilize scatter plots in Figs. 1, 2, and 3 across combinations of predictors and the response variable to investigate for potential complete separation in the data. Since the scatter plots are colored based on the $Y = 0$ and $Y = 1$ classes, we can see if the classes in the response variable are separated according to the pairs of independent features in these scatter plots. Different combinations of predictors are analyzed to assess which predictors perfectly separate the binary response variable, Class, indicating whether the transaction is fraudulent or not. Investigating the first set of predictor combinations, as shown in Fig. 1, there is a distinct separation in the binary response variable for V3–V4, V7–V8, V9–V10, and V11–V12 predictors. This is shown by the positive class instances clustering away from the negative class. Fig. 2 shows more distinct clustering for V13–14, V15–16, and V17–18. Fig. 3 shows no obvious clustering for any of the final three combinations. Thus, we can conclude that significant complete separation is present for at least half of the combinations of predictors on the binary response variable.

To assess the co-relationship patterns between the response variable and predictors, we calculate the point bi-serial correlation between the

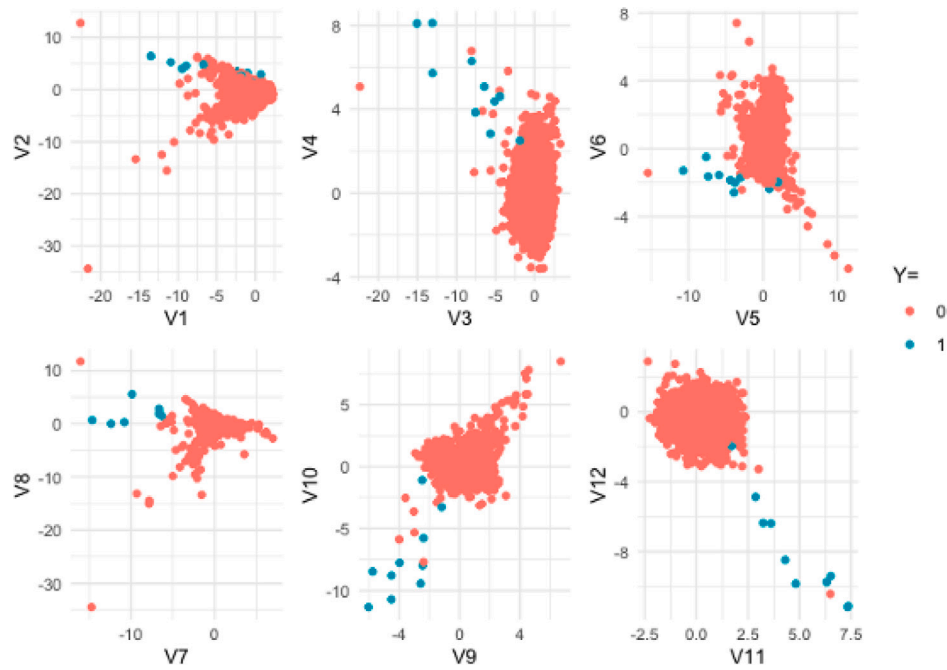


Fig. 1. Scatter plots of predictors and response variable for separation detection for the first 12 predictors.

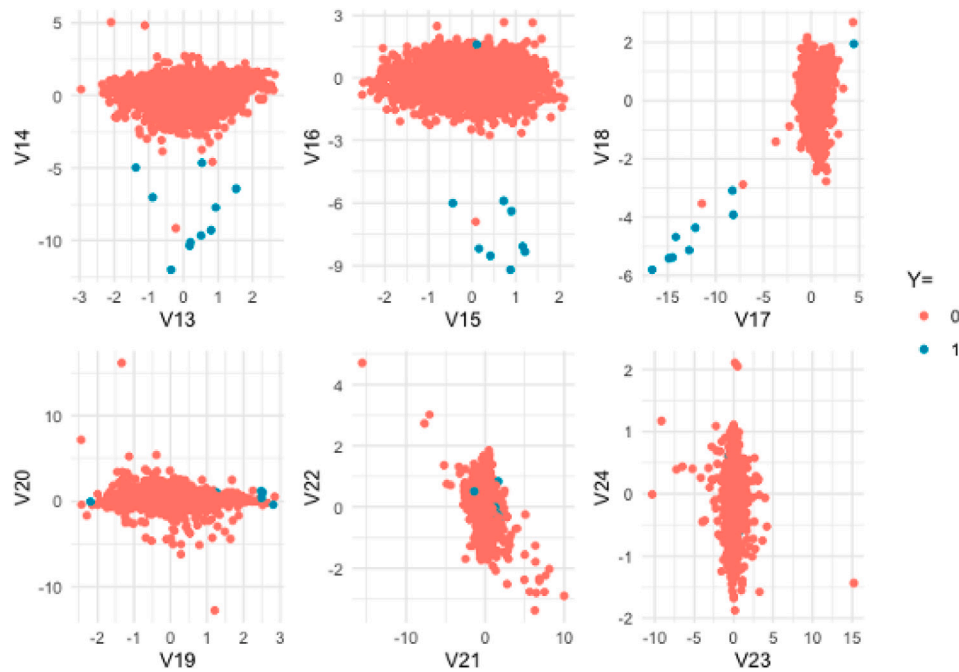


Fig. 2. Scatter plots of predictors and response variable for separation detection, for the second set of 12 predictors.

pairs of binary responses and continuous predictors and the Pearson correlations between the pairs of predictors. The results are shown in Fig. 4. There is no notable correlation between the pairs of predictors except those with the transaction amount. This is a desired result to avoid multicollinearity. Most predictors are significantly correlated with the response variable at a 5% significance level, as shown by an “*” next to the predictor name in Fig. 4. Thus, the predictors are associated with the response variable and can contribute to the model for classification.

Fraudster behavior and consumer demographics/characteristics cause patterns to vary in the data significantly and rapidly. To examine the level of changes in the data characteristics, we employ the time

series plots and the coefficient of variation (CV) to assess the variation in the predictors since the measurement units of the predictors are masked. CV is defined as the ratio of standard deviation to the mean of a variable. Table 1 shows the CVs of the predictors. All predictors, except the amount of fraudulent activity (V29), exhibit extremely large variation, implying a high level of change in each predictor. The time series plots of predictors in Fig. 5 appear to be centered around zero. However, they have rapid spikes and sudden changes in the range of the series, indicating high volatility. Thus, sudden changes in the characteristics of the fraud dataset are evident from CVs and time series plots of predictors.

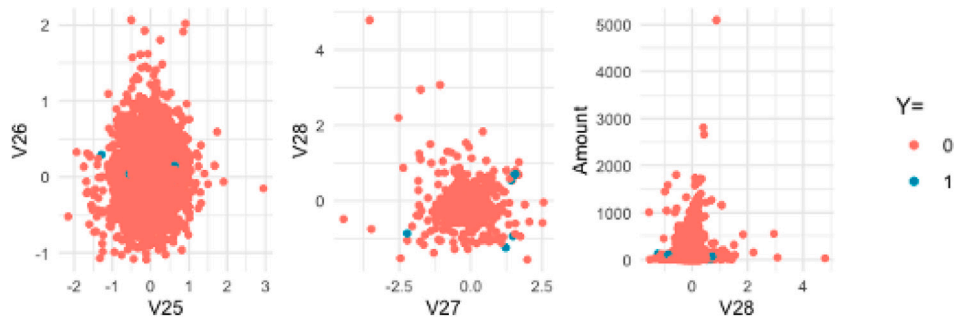


Fig. 3. Scatter plots of predictors and response variable for separation detection for the final set of 6 predictors.

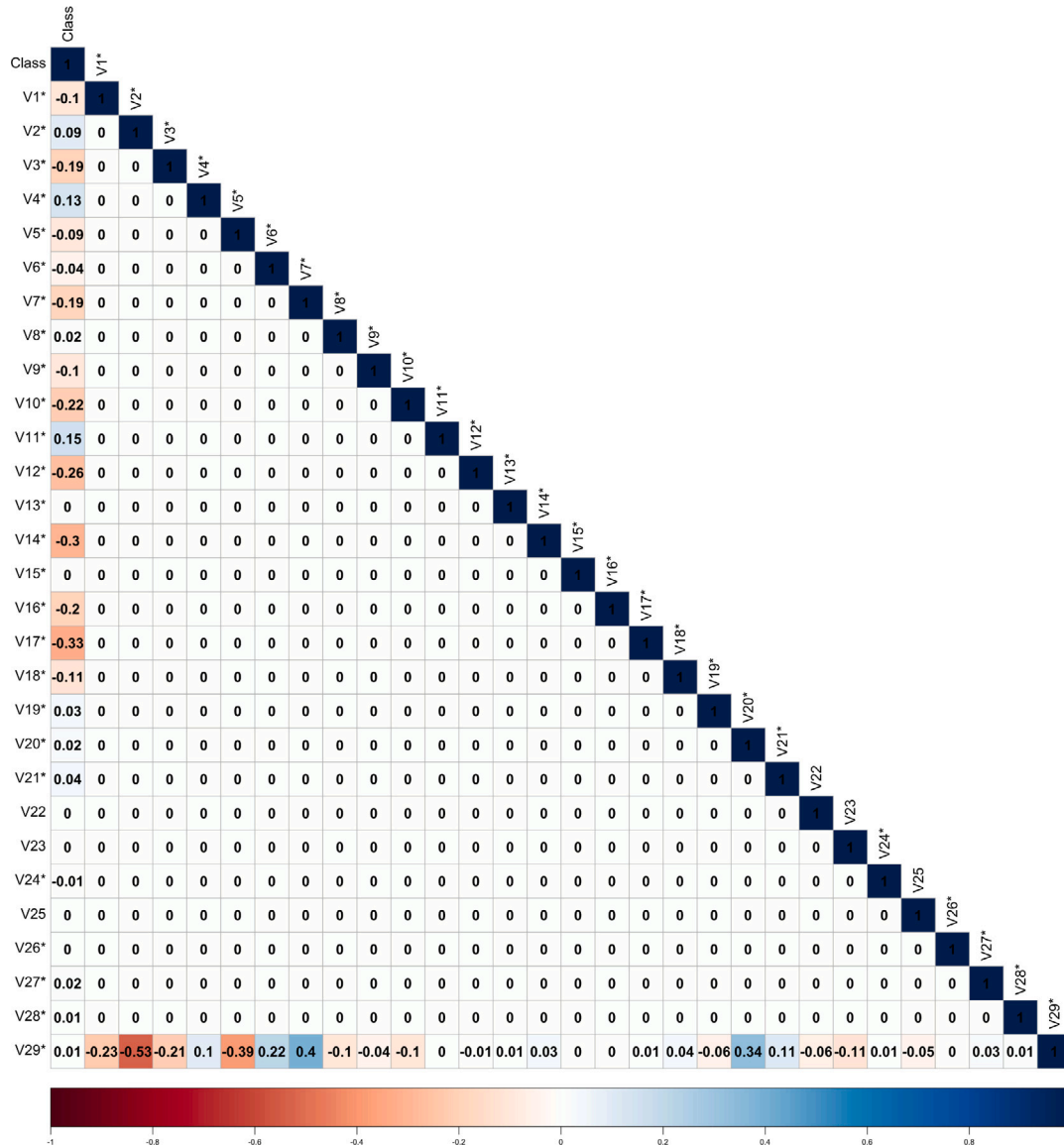


Fig. 4. Point bi-series correlations between the pairs of the response variable and predictors and the Pearson correlations between the pairs of predictors. The value 0 in cells represents a correlation less than 0.01. An '*' next to the variable name indicates the significance of the correlation between the response variable and the corresponding predictor.

2.2. Simulated transaction fraud dataset

To further enhance the generalizability of the proposed online system using fuzzy logistic regression, we consider an additional simulated dataset by Shenoy and Harris (2021). This dataset includes transaction data across 800 merchants and 1000 customers during a six-month

period from June 2020 to January 2021. This simulated dataset is fairly extensive in the information included for each transaction, aiming to represent a real-world scenario. It includes information about customer demographics, such as age, location of residence, and job description, along with full details of merchant facilities. There are 555,000 transactions, all occurring within the US, with a rate of imbalance at 99.6%

Table 1
Coefficient of variation values of all predictors.

V1	V2	V3	V4	V5	V6	V7	V8	V9	V10
1.1E+12	-2.0E+12	-1.6E+12	1.7E+12	8.4E+12	3.1E+12	-4.0E+12	1.4E+13	-9.3E+11	1.5E+12
V11	V12	V13	V14	V15	V16	V17	V18	V19	V20
5.4E+11	9.5E+11	1.4E+12	-6.5E+12	-1.7E+12	-3.8E+12	-1.3E+12	1.7E+12	1.2E+12	4.4E+11
V21	V22	V23	V24	V25	V26	V27	V28	V29	
-2.2E+12	-1.3E+12	-6.4E+11	4.1E+11	-7.5E+11	-8.6E+11	1.2E+11	-9.4E+10	2.8E+00	

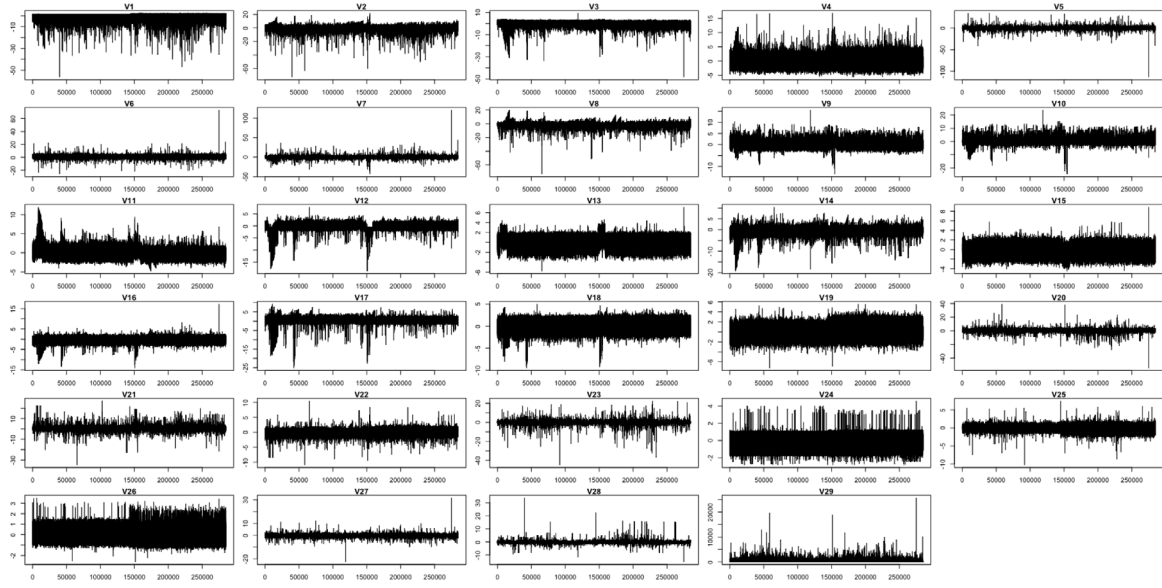


Fig. 5. Time series plots of predictors to examine changing fraud characteristics.

towards the $Y = 0$ class. Only 0.39% of transactions are actually fraudulent. We utilize 11 predictors, which include merchant type, category of products, amount of transaction, customer city location, state in the US where the customer is located, the population of the city in which the customer is located, their job, age, time of the transaction occurrence, and the merchant's coordinates with some located overseas.

Scatter plots in Fig. 1, 2, and 3 of the Supplementary Material are created to examine complete separation in the data using sets of predictors and binary responses. Using the scatter plots, we look for distinct clustering patterns in the binary responses for any set of predictors. Out of 16 considered combinations of predictors, 9 show distinct clustering of the binary responses. More specifically, Unix time and merchant location, type, product category, amount, city and state of customer's residence, population demographic, and customer's age showcase strong clustering of the binary responses, providing evidence for complete separation in the data.

2.3. Fuzzy logistic regression

The fuzzy logistic regression model, defined in Eq. (1), captures the relationship between the fuzzy binary response variable, $\bar{Y}$, and a set of predictors, $\mathbf{X} = (X_i), i = 1, \dots, k$, by using the associated fuzzy coefficients, $\bar{\beta} = (\bar{\beta}_j), j = 0, 1, \dots, k$, and fuzzy success probabilities, $\bar{p} \in [0, 1]$.

$$\log\left(\frac{\bar{p}}{1-\bar{p}}\right) = \bar{\beta}_0 + \bar{\beta}_1 X_1 + \bar{\beta}_2 X_2 + \dots + \bar{\beta}_k X_k \quad (1)$$

2.3.1. Fuzzification of binary responses

A binary triangular fuzzy number (TFN), $\bar{Y}$, defined by three values $y_1 < y_2 < y_3$, and a membership function $\mu_Y(x)$ (Dubois & Prade, 1978), operates as a mapping from a binary discrete set $\mathbb{Z}_2 = \{0, 1\}$ to $\mathbb{R}' = \{x \in \mathbb{R} | -1 \leq x \leq 1\}$ and is a function $F : \mathbb{Z}_2 \rightarrow \mathbb{R}'$ defined as

$F(Y) = [Y - m \cdot \ell \cdot U, Y, Y + m \cdot r \cdot U]$, where $U \sim \text{Uniform}(I_L, I_U)$ and m is the degree of fuzziness. Here r and ℓ are to adjust the symmetry of output TFN. The upper and lower limits, I_L and I_U are user-specified parameters (Charizanos et al., 2023).

2.3.2. Random fuzzy vectors for coefficient estimates

Random fuzzy vectors $\bar{V}_k = (\bar{V}_{1k}, \dots, \bar{V}_{Nk})$ are generated within a pre-defined range of $\bar{I}_\beta = [\bar{I}^-, \bar{I}^+]$ by ordering random crisp vectors, such that $x_{1k} < x_{2k} < x_{3k}$, from a set of randomly generated crisp vectors, $v_k = (x_{1k}, \dots, x_{3N+3,k})$, where $x_{ik} \in [0, 1]$. These fuzzy vectors are used for the generation of fuzzy coefficients, $\bar{\beta}$. Then, $\bar{\beta}$ are plugged in the model equation in Eq. (1), and fuzzy success probabilities $\bar{P}$ are computed. The process is repeated M times, defined by the user as the number of Monte Carlo (MC) replications. Optimization is achieved by the use of the fuzzy definitions for MSE, MAE, or RMSE optimization measures, while Specificity, Sensitivity, F1 Score, and MCC are computed as performance measures (Charizanos et al., 2023).

2.3.3. Classification of subjects with a fuzzy threshold

The vertex position of each TFN probability, $\bar{P}$, is classified by comparing against a fuzzy threshold $\bar{\tau} = (a_1, a_2, a_3)$. Firstly, the expected value of $\bar{P}$, $\mathbb{E}(\bar{P})$, is calculated following Dubois and Prade (1978). The difference between $\bar{P}$ and $\bar{\tau}$ and width, $w(\mathbb{D})$ are estimated. If $\mathbb{E}(\bar{P}) < a_2$, $\hat{y}_2 = 0$, otherwise, $\hat{y}_2 = 1$. The outcome of this process is a matrix representing the output $\hat{Y}$ where each element is a triangular fuzzy number with the vertex $\hat{y}_2$ is equal to either 0 or 1, while $\hat{y}_1$ and $\hat{y}_3$ points are estimated by $\hat{y}_1 = \hat{y}_2 - w(\mathbb{D})$ and $\hat{y}_3 = \hat{y}_2 + w(\mathbb{D})$.

2.4. Real-time fuzzy fraud detection

This section presents the proposed fraud detection framework, starting with a general description and delving into the implementation

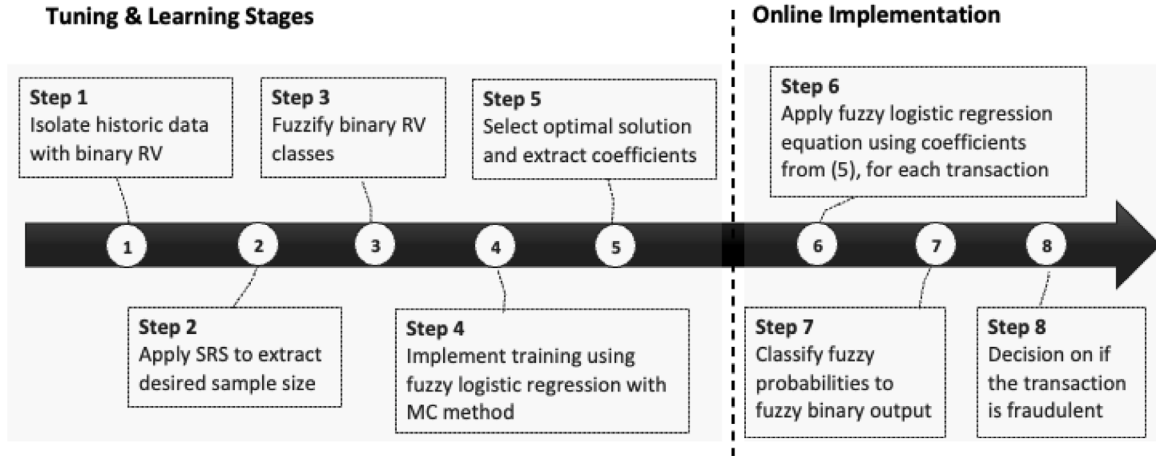


Fig. 6. General implementation of credit card fraud detection with binary response variable (RV), using fuzzy methodology and stratified random sampling (SRS) to handle non-stationary changes in transaction data.

details. A general outline of the methodology is first given. Then, we explore the handling of non-stationary changes in transaction fraud characteristics, the tuning and learning stages of the fuzzy logistic framework, and finally, the online implementation using fuzzy coefficients.

2.4.1. Outline of implementation

Fig. 6 displays the overall process of implementing our methodology for transaction fraud detection. The first step is to gather all the required historical data, with required binary responses for fraud and non-fraud transactions, along with the predictors. The volume of the historical data can be large. This does not create any issues in our framework. Step 2 involves using stratified random sampling to handle non-stationary changes in fraud characteristics, as defined in Section 2.4.2. In Step 3, the binary responses are fuzzified using the method outlined in Section B.1, and then the tuning and learning stages are implemented in Step 4. In Step 5, we extract the fuzzy coefficients using the single optimal solution from the tuning and learning stages. These are then applied to online transactions using the fuzzy logistic regression equation for each transaction at a time. We then classify the fuzzy probabilities that are output from Step 6 into fuzzy binary outcomes, which are used in the final step to decide whether a transaction is fraudulent or not, taking relevant action to protect consumer accounts. The methodology as a whole is executed with R, version 4.2.3. Our codes for the method and applications in this article are given at <https://github.com/GZanos/FFL>. Fig. 7 presents details and the flow chart of the proposed framework with every function defined under Sections 2.3 and 2.4.

2.4.2. Handling non-stationary changes in fraud transaction data

Continuous changes in fraud characteristics govern the transaction fraud environment. The types of transaction fraud can change suddenly and frequently. This creates a highly sensitive environment towards non-stationary changes and the necessity to handle such patterns in the data. In our methodology, stratified random sampling (Cochran, 1977; Kalton & Flores-Cervantes, 2003) is used to apply weights in the transaction data using a timestamp component. In this way, we assign a higher representation to the recent fraudulent transactions than the older ones while keeping the level of randomness stable with a probabilistic sampling technique (Cochran, 1977).

The process to handle the data for the proposed fraud detection framework is outlined in Fig. 8. The dataset of $N + V$ transactions that will be used to train the fraud detection system is first divided into training and validation portions of sizes N and V without changing the order of transactions. Thus, a proportion of the transactions at the last segment of the $N + V$ transactions is separated as the validation

portion. Random sampling must be avoided to preserve the changing fraud and consumer characteristics while specifying the validation set.

The stratified random sampling is implemented on the training portion of the whole dataset. The overall population of N credit card transactions in the training portion is first divided into two strata: S_1 and S_2 , given a timestamp threshold t . This timestamp threshold is user-defined and represents the desired timestamp that separates the periods of recent and historical data. Random sampling is then used to create two new samples s_1 and s_2 of sizes n_1 and n_2 from S_1 and S_2 , respectively. n_1 and n_2 are specified based on the characteristics of the fraud dataset and the availability of computational resources. The same rate of $Y = 1$ class is maintained in both s_1 and s_2 . We analyze the within-stratum variance, $V_i, i = 1, 2$, for both recent and historical data using Eq. (2). A lower within-stratum variance indicates less variability, hence a more homogeneous stratum that better represents non-stationary changes in recent fraud characteristics.

$$V_i = \left(\frac{1}{n_i} \sum_{j=1}^{n_i} I\{Y_i \in S_i\} \right) \left(1 - \frac{1}{n_i} \sum_{j=1}^{n_i} I\{Y_i \in S_i\} \right), \quad (2)$$

where $V_i \in [0, 1]$ and $I\{\cdot\}$ is an indicator function. Then, we find the weight factor towards recent data in the first stratum relative to both strata using Eq. (3):

$$W_t = \frac{V_1}{V_1 + V_2} \quad (3)$$

We take $n_1 W_t$ observations from s_1 and $n_2 W_t$ from s_2 to create the training sample. The training sample is then used for hyper-parameter tuning and learning stages.

2.4.3. Tuning and learning stages

The tuning stage aims to find the optimal parameter settings under which the model showcases the best performance results for the fuzzy logistic regression model of Section 2. There are several user-specified parameters in the fuzzy logistic regression model. These include the limits for generating random coefficient values, degree of fuzziness m , asymmetry points, l, r , and minimum and maximum fuzzification limits. Users can also select different fuzzy thresholds for classification. The model is implemented over a grid of different parameter settings. Performance and optimization results are extracted for each combination of the parameters in the grid, which allows for the assessment of how the framework performs under different conditions. This performance analysis allows for selecting the single optimal set of user-specified parameters regarding the model's performance. It also helps understand the performance and optimization sensitivity of the model towards important parameter settings, such as fuzzification and asymmetry, or choosing different thresholds for classification.

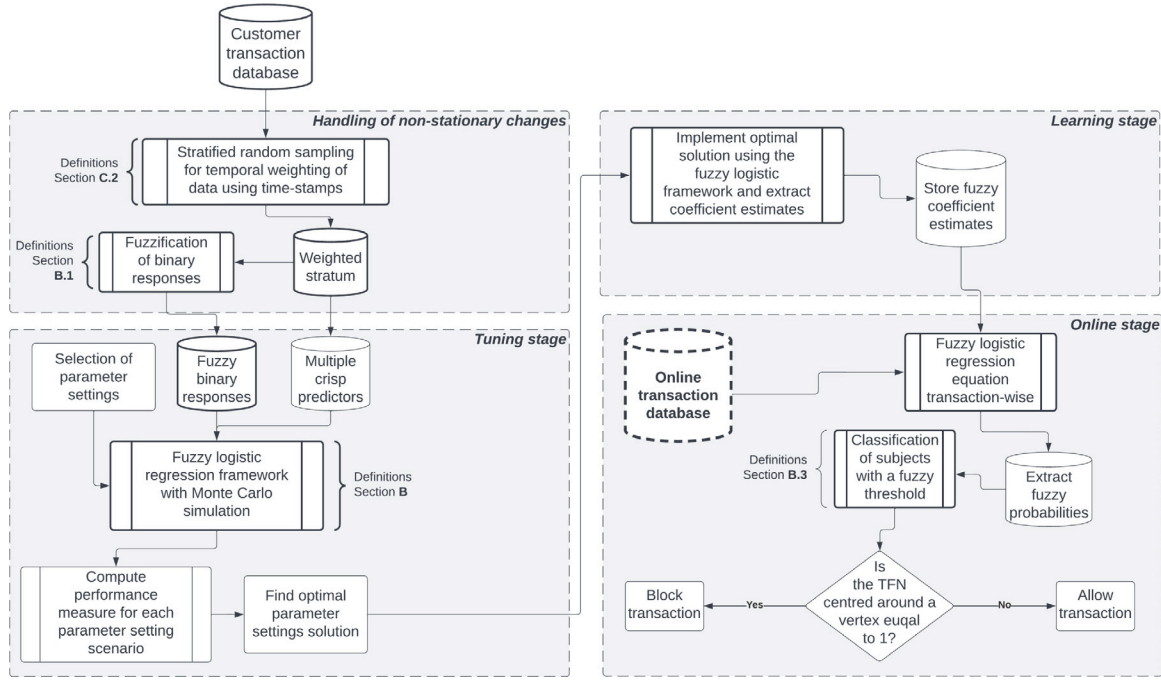


Fig. 7. Flowchart of the proposed fraud detection framework.

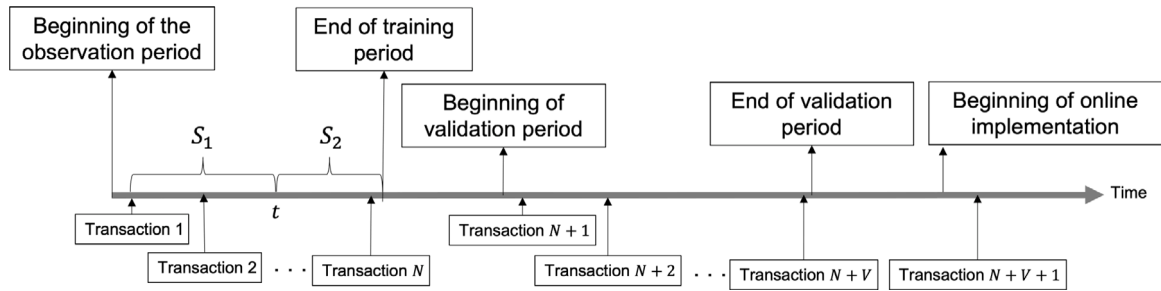


Fig. 8. Handling a fraud dataset to build the fraud detection framework.

The learning stage is then achieved by implementing the framework once more on the optimal parameter settings determined after the tuning stage. On this last implementation, performance is assessed once more, and fuzzy coefficient estimates are extracted. These are stored separately to be used in the online implementation. Alternatively, the coefficient estimates can be extracted from the run with the optimal set of parameters if the coefficient estimates are recorded for each run at the tuning stage.

Before the first online implementation of the fraud detection system, a validation effort needs to be conducted to confirm the usefulness of the parameter estimates. V transactions in the validation dataset are sent to the trained model sequentially to mimic the model's online use and validate the trained model's classification performance. The size of the validation dataset needs to be large enough to observe a consistent picture of the model's performance.

2.4.4. Online implementation

In the online implementation stage, the new data come from the fraud monitoring system after the tuning and learning stages. This is represented by the "Online transaction database" in the flow chart in Fig. 7. The new data must have the same features as in the historical dataset used in the tuning and learning stages.

The fuzzy coefficient estimates $\bar{\beta}_0$ and $\bar{\beta}_i$, extracted during the learning stage, together with predictor information, are applied to each new transaction coming live one by one using Eq. (1). This instantly

generates fuzzy probability estimates, which are classified using a fuzzy threshold $\bar{\tau} = (a_1, a_2, a_3)$ to create predicted outcomes in the form of $\bar{Y} = (\hat{y}_1, \hat{y}_2, \hat{y}_3)$. Finally, a decision is made based on the shape of each predicted TFN and vertex position. TFN predictions with a vertex position of 1 indicate potentially a fraudulent transaction that must be blocked. A vertex of 0 indicates a non-fraudulent transaction that can be allowed to occur. If the new transaction is classified as fraudulent, a notification is created for the relevant team to block the transaction and notify the customer. The customer may receive a call from the relevant banking department advising them of this activity and taking steps to prevent it in future, which may involve the cancellation and re-issue of credit card account numbers. If the transaction is not classified as fraudulent, it is simply allowed to occur.

3. Application with real transaction data

3.1. Performance against sample size and Monte Carlo replications

We investigate the impact of sample size and the number of MC replications on efficiency and performance to determine if the fuzzy logistic regression model's performance is sensitive to both sample size and the number of MC replications. Given our study's key objective, this analysis becomes imperative in dealing with the efficiency versus performance nexus.

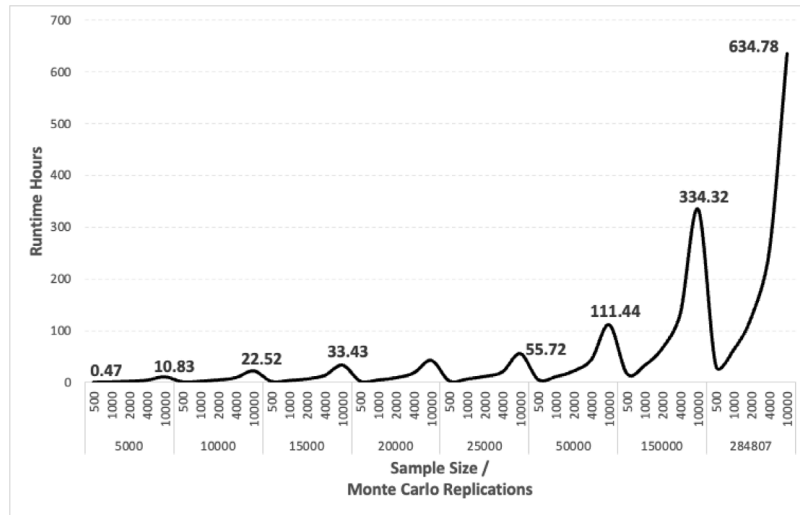


Fig. 9. Runtime hours per sample size (vertical numbers) and Monte Carlo replications (horizontal numbers).

For this analysis, we use simple random sampling to extract different sample sizes, ensuring the same rate of $Y = 1$ class (0.2%) is kept in each sample. The smallest sample is $n = 5000$, and the largest is $n = 150,000$. The number of MC replications is a user-selected parameter in the fuzzy logistic regression model. We test the number of MC replications from 500 to 10,000. These experiments were done on a computer with Apple MAC M2, 3.5 GHz CPU, and 8 GB of memory.

Fig. 9 shows the computation time of each run based on sample size and the number of MC replications. The results show that we should expect a sample size of $n = 5000$ with only 500 MC replications to be associated with 0.47 h of run time. This increases to 10.83 h for the same sample size but with 10,000 MC replications. Run time then starts to increase exponentially as we induce larger samples with the same range of MC replications. For $n = 10,000$, average run time increases by 107%, while $n = 15,000$ by a further 48%. The highest estimated run time is 634.78 h (26.5 days) for the total population of $n = 284,807$ and $MC = 10,000$ replication.

Fig. 10 shows the performance results for each sample size and number of MC replications. Performance measures are defined in Table 1 of Supplementary Material. Performance scores across specificity, sensitivity, F1 scores, and MCC do not show any significant variation as sample size and number of MC replications increase. Sensitivity remains high at 0.996, whilst specificity is greater than 0.856 for all selections. The sensitivity score indicates that irrespective of sample size and the number of MC replications within the considered limits, the model correctly identifies 99.6% of transactions classified as fraudulent. It also correctly identifies 85.6% of actual negative classes, as per the specificity score. These results suggest that even with a much smaller sample and number of MC replications, the model is highly capable of correctly identifying both fraudulent and normal transactions. F1 scores exceed 0.796 and MCC at 0.633 or above. These results indicate that the model achieves a good balance between precision and recall, while the model's predictions are better than random. Furthermore, this performance is almost identical for all sample sizes and the number of MC replications.

We conclude that the fuzzy logistic regression framework performs equally well with a smaller sample size and number of MC replications. This allows for selecting $n = 5000$ with just $MC = 500$ replications as the identical solution to achieve optimal results with greater efficiency in the tuning stage. The ability to work on smaller sample sizes and fewer MC replications allows for much greater efficiency for the fraud detection framework overall. This is because the tuning and learning stages can now finish in a matter of hours rather than days or weeks if we had to work with the complete sample. Given that performance

Table 2

Hyper-parameter tuning settings.

Tuning range	1	2	3	4	5
β_0	[-10,10]	[-5,5]	[-2,2]	[-1,1]	[-0.5,0.5]
β_1	[-10,10]	[-5,5]	[-2,2]	[-1,1]	[-0.5,0.5]
Parameter	Values/Intervals selected				
Fuzzification factor	$m = (1, 1.5, 2.5, 4.5)$				
Asymmetry setting	$(l, r) = [(4.5, 1.5), (1.5, 4.5), (1, 1), (2.5, 0.5), (0.5, 2.5)]$				
Binary fuzzification limits	$I_L = 0, I_U = (0.5, 1.5)$				
Fuzzy threshold	$\bar{\tau}_1 = (0.4, 0.5, 0.6)$				

Table 3

Optimization performance statistics for MSE, MAE, and RMSE measures.

Opt. Measure	Lowest value	Highest value	Mean	Std. Dev.	CV
MSE	0.122	315.54	30.533	68.015	2.228
MAE	0.383	19.011	3.993	4.348	1.089
RMSE	0.211	9.157	1.929	2.093	1.085

is not impacted due to the reduction of sample size or the number of MC replications, we can achieve optimal solutions with a similar level of performance quicker.

3.2. Results of tuning and learning stages

We first perform a parameter tuning using the ranges supplied in Table 2 and 500 MC replications to tune up the model and find the optimal parameter settings under which the strongest performance and optimization are achieved.

Table 3 shows the results for each optimization measure at the tuning stage. We calculate the coefficient of variation (CV) as a measure of variability, along with the minimum and maximum error estimates. CV values show that MAE and RMSE have similar levels of variability. On the other hand, MSE produces a 104.9% larger CV score compared to MAE and RMSE. This indicates that for MSE-optimized models, there is substantially higher variability in optimization performance, hence, more sensitive results.

Fig. 11 shows the distribution of error estimates for each optimization measure. The box plots in Fig. 11 have fewer outliers for MAE and RMSE, with only one outlier for MAE and two for RMSE. In comparison, MSE produces three times as many outlier values. Given the lowest number of outliers, a more consistent distribution of scores, and a low CV value, MAE is chosen as the optimal error measure for model optimization.

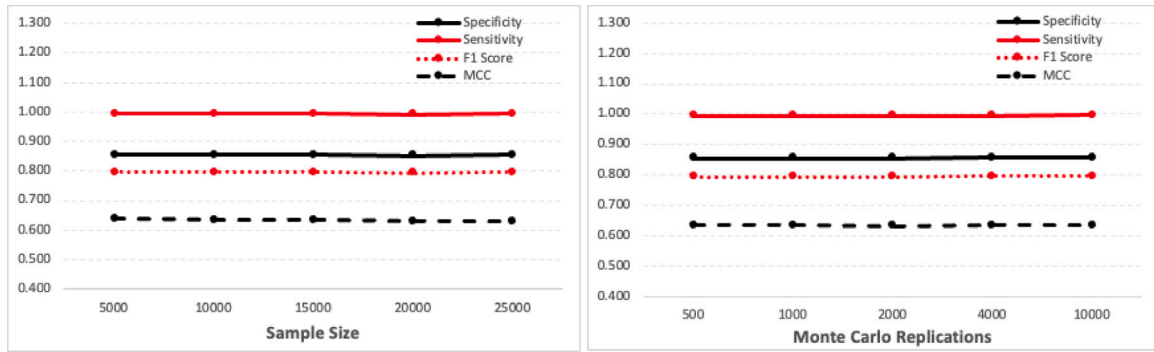


Fig. 10. Performance results based on sample size and the number of Monte Carlo replications.

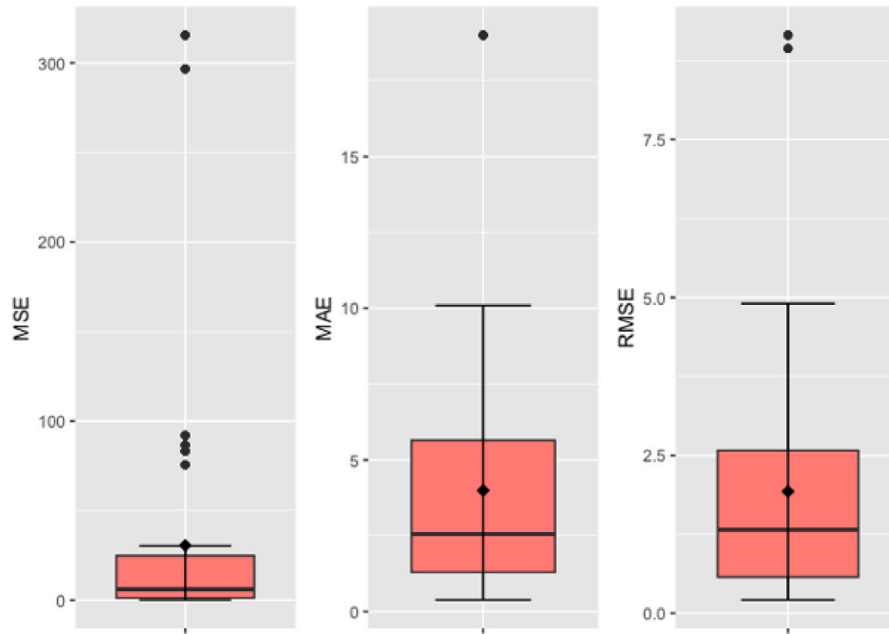


Fig. 11. Box-plot distributions for optimization measures across 200 tuning runs.

Using MAE as the optimization measure, the optimal parameter settings are found by ranking the combinations of settings for best performance scores across all performance measures and selecting the top setting as the single best solution. The complete analysis of tune-up tests can be found in Section A of the Supplementary Material. The optimal parameter settings, following the tune-up analysis, are as follows. Degree of fuzziness for binary fuzzification set at a high level $m = 1.5$. The asymmetry is set at minimum $l = 1, r = 1$. Coefficient range for β_0 and β_i is set at $[-1, 1]$. The fuzzy threshold is set to $(0.4, 0.5, 0.6)$. Binary fuzzification minimum and maximum limits are set to $[0, 0.5]$. MAE is used as the optimization measure.

We implement the fuzzy logistic regression model with the above optimal settings to achieve the learning stage for the model. The performance results are shown in Table 4. Specificity and sensitivity scores are high, 0.902 and 0.997, respectively. The model correctly identifies 90.2% of non-fraudulent and 99.7% of fraudulent transactions. The F1 score of 0.878 suggests a good balance between precision and recall in classifying fraudulent transactions. Lastly, the MCC of 0.762 reflects the strong quality of the model's predictions. Overall, these results indicate that this optimal set of parameters provides a solid performance for the model at the learning stage.

Considering the 99.8% imbalance towards the negative class ($Y = 0$), a model influenced by the imbalance issue would show high specificity and low sensitivity scores. F1 and MCC would also be low. Not

observing any of these indicators, we can conclude that the model is not impacted by imbalance. Furthermore, the lack of perfect scores across all performance measures indicates that complete separation does not impact model performance.

Fuzzy coefficient estimates shown in Table 5 are extracted to be used in the online implementation stage for real-time fraud detection. Table 5 also depicts defuzzified values of coefficient estimates and their exponential values (odds ratios) to assess the predictors' contribution to credit card detection classification. This information also helps understand the reasons why an operation became suspicious. Most predictors have an odds ratio (or its reciprocal) notably greater than one. Although V25 has a slightly insignificant correlation (Fig. 4, P -value = 0.07) with the response variable, it has the highest contribution according to the odds ratios. A one-unit change in V25 increases the odds of being a legitimate transaction by 2.17 times. Thus, V25 is the most influential predictor for labeling a transaction as suspicious. V25 is followed by V10, V1, and V18. A one-unit change in these predictors increases the odds of being a legitimate transaction by at least 1.5 times. On the other hand, a one-unit change in V26 increases the odds of a fraudulent transfer by 1.82 times. A one-unit change in V12, V16, and V13 increases the odds of being a fraudulent transaction by at least 1.5 times. In the case of a real-world banking application, these predictors should be monitored closely to take preventive measures.

Table 4

Performance results of the optimal solution in the learning stage when MSE, MAE, or RMSE is used as the optimization measure.

Performance measure	MSE	MAE	RMSE
Opt. Error estimate	1.385	1.366	0.615
Specificity	0.902	0.902	0.902
Sensitivity	0.997	0.997	0.998
F1 Score	0.878	0.878	0.878
MCC	0.763	0.762	0.763

3.3. Validation and online implementation

To examine the performance of our framework in an online environment, we implement the fuzzy logistic regression model one by one on a test set of $V = 5000$ transactions to mimic the processing of 5000 new transactions. These were initially extracted as the most recent transactions list from the complete data to mimic an online environment. The time required to process 5000 transactions takes 8.4 min in a simulated online environment with the computer mentioned in Section 3.1. This corresponds to 0.1 s to process a new transaction in an online environment.

The TFN coefficient estimates in Table 5 are plugged in Eq. (1) along with the values of predictors for each transaction at a time. Each transaction is evaluated in 0.0056 s on average, indicating that a prediction for each transaction is efficiently generated on a real-time basis. Each implementation of the model for each transaction produces a fuzzy probability prediction. This fuzzy probability is then classified using the method under Section 2.4.3, and a fuzzy prediction is drawn in the form of a triangular fuzzy number. The vertex position of each TFN prediction has the values of either 0 or 1. Following the methodology defined in Section 2.4.4, a decision is made on whether to block or allow each transaction.

After completing the processing of 5000 transactions, the confusion matrix given Table 6 is obtained to assess classification performance in the online environment. Out of 5000 transactions, we know that 10 transactions are fraudulent and 4990 are non-fraudulent. The model correctly classifies 4962 out of 4990 non-fraudulent transactions and 10 out of 10 fraudulent transactions. At the same time, we also have one transaction that is incorrectly identified as non-fraudulent (false negative) and 17 transactions that are incorrectly identified as fraudulent (false positive). The sensitivity, specificity, F1 score, and MCC are obtained as given in the first row of Table 7. The online stage of the fuzzy logistic regression framework shows strong specificity and sensitivity scores of 0.995 and 0.909, respectively. An F1 score of 0.526 indicates a good balance between precision and recall, while MCC at 0.809 suggests a balanced and better than random chance classification performance. These are balanced results across all performance measures as desired.

3.4. Use of other ML methods in the proposed framework

We further implement five crisp ML methods to use as a benchmark against the fuzzy logistic regression framework. These are Logistic Regression, K-Nearest Neighbors (KNN), Naive Bayes, Artificial Neural Network, and XG-Boost. All crisp ML methods were hyper-parameter-tuned and trained on the same weighted stratum, also used for the fuzzy logistic regression framework. Testing is also done using the same online transactions, which were separated for testing purposes of the fuzzy logistic framework as well. Following this approach, we simply use a crisp ML model instead of the fuzzy logistic regression in our fraud detection framework. Thus, the proposed framework and the set of crisp ML methods are implemented on the same datasets to ensure comparable results.

Table 7 shows the classification performance results of the online implementation for the proposed framework and the benchmark ML

methods. As discussed, the online implementation of the fuzzy logistic regression framework has high performance in detecting both fraudulent and non-fraudulent transactions. On the other hand, all five crisp ML methods show extremely high specificity and low sensitivity scores, indicating that these ML methods performed well in identifying non-fraudulent transactions but poorly in flagging fraudulent transactions. Such performance is indicative of influence from strong imbalance towards $Y = 0$. F1 scores and MCC values remain relatively low for these methods, providing further evidence that ML methods do not work well, given the level of imbalance in the data. However, given the higher sensitivity scores, fuzzy logistic regression handles this imbalance well.

3.5. Comparison with other fraud detection methodologies

We compare the performance of our framework against other recently published fraud detection approaches. Table 8 shows the reported performance measures of these approaches. Since not all the considered studies report as many performance measures as our study, the corresponding cells of Table 8 are left blank.

One method for detecting fraud anomalies in transaction data is a hybrid fuzzy approach that utilizes fuzzy clustering techniques with a neural network (Fuzzy Clustering - ANN) (Behera & Panigrahi, 2015). The fuzzy clustering part is used in the behavioral analysis phase, with a neural network model in the learning stage. This methodology utilizes the same dataset as in our work. However, the data is divided into 70% for training, 15% for validation, and 15% for testing. The training data is used to train the model, the validation data is used to evaluate the model and hyperparameter tuning, and the test data is used to evaluate the final online model. The analysis of results using ROC curves indicates that fraudulent transactions have been classified with 93.9% accuracy on the test dataset. Our framework achieves a fraud detection rate of 90.9%, indicating a similar sensitivity level in classifying fraudulent transactions. However, Behera and Panigrahi (2015) do not mention the accuracy of non-fraudulent transactions (i.e., negative class). If both specificity and sensitivity were computed and analyzed, more targeted insights could be derived to understand the impact of imbalance. Furthermore, our framework handles the imbalance and complete separation issues without impact on performance. This is shown by investigating additional performance measures, such as F1 and MCC, achieving a more holistic understanding of model performance.

Moreover, an ML model that utilizes a rule induction in a distributed tree-based algorithm (Distributed Tree-based) is implemented to detect fraudulent transactions (Vorobyev & Krivitskaya, 2022). This approach extracts interpretable rules learned by the tree-based ML method to describe the fraudulent activity for classification. The mlr3 R package is used to access the tree-based algorithm TaskClassif, which allows for replicating Vorobyev and Krivitskaya's (Vorobyev & Krivitskaya, 2022) analysis. To implement Vorobyev and Krivitskaya's (Vorobyev & Krivitskaya, 2022) approach with our dataset, we used 70% of the whole data to train the tree-based model and the remaining 30% for testing. The performance of this model shows a sensitivity score of 0.760, an almost perfect specificity of 0.999, F1 score of 0.527, and MCC of 0.830. These results indicate that 76% of fraudulent and almost 100% of non-fraudulent transactions are classified correctly. This variation between sensitivity and specificity scores, with specificity almost a perfect score of 1.000 and much lower sensitivity, indicates that the class imbalance impacts the model performance. However, we see good F1 and MCC results, indicating a good balance between precision and recall. In comparison, our framework shows a much higher sensitivity score and a more balanced specificity score, indicating no impact on model performance from class imbalance. Furthermore, the granular results of the fuzzy logistic regression framework have 17 false positives and only one false negative. In contrast, the tree-based algorithm has

Table 5

TFN coefficient (Coef.) estimates, defuzzified crisp values of TFN coefficients, and their exponential values (Exp.) obtained in the learning stage.

Coef.	TFN estimate	Crisp value	Exp.	Coef.	TFN estimate	Crisp value	Exp.
$\hat{\beta}_0$	(−0.920, −0.587, 0.552)	−0.652	0.521	$\hat{\beta}_{15}$	(−0.082, 0.433, 0.529)	0.293	1.340
$\hat{\beta}_1$	(−0.699, −0.458, −0.314)	−0.490	0.613	$\hat{\beta}_{16}$	(0.199, 0.482, 0.854)	0.512	1.669
$\hat{\beta}_2$	(−0.963, 0.345, 0.638)	0.007	1.007	$\hat{\beta}_{17}$	(−0.671, −0.095, 0.184)	−0.194	0.824
$\hat{\beta}_3$	(0.143, 0.316, 0.503)	0.321	1.379	$\hat{\beta}_{18}$	(−0.913, −0.443, 0.047)	−0.436	0.647
$\hat{\beta}_4$	(−0.889, −0.518, 0.664)	−0.248	0.780	$\hat{\beta}_{19}$	(−0.932, 0.068, 0.493)	−0.124	0.883
$\hat{\beta}_5$	(−0.611, −0.504, 0.256)	−0.286	0.751	$\hat{\beta}_{20}$	(−0.955, −0.336, 0.161)	−0.376	0.687
$\hat{\beta}_6$	(−0.229, 0.105, 0.837)	0.238	1.269	$\hat{\beta}_{21}$	(−0.605, −0.254, −0.018)	−0.292	0.747
$\hat{\beta}_7$	(0.017, 0.397, 0.597)	0.337	1.401	$\hat{\beta}_{22}$	(−0.591, −0.534, 0.582)	−0.181	0.834
$\hat{\beta}_8$	(−0.455, 0.223, 0.778)	0.182	1.200	$\hat{\beta}_{23}$	(−0.651, −0.366, 0.508)	−0.170	0.844
$\hat{\beta}_9$	(−0.533, −0.071, 0.895)	0.097	1.102	$\hat{\beta}_{24}$	(−0.260, 0.363, 0.425)	0.176	1.192
$\hat{\beta}_{10}$	(−0.954, −0.654, 0.047)	−0.520	0.595	$\hat{\beta}_{25}$	(−0.836, −0.820, −0.672)	−0.776	0.460
$\hat{\beta}_{11}$	(−0.780, −0.169, 0.693)	−0.085	0.919	$\hat{\beta}_{26}$	(0.289, 0.581, 0.931)	0.600	1.822
$\hat{\beta}_{12}$	(0.248, 0.477, 0.984)	0.570	1.768	$\hat{\beta}_{27}$	(−0.520, −0.176, 0.259)	−0.146	0.864
$\hat{\beta}_{13}$	(−0.073, 0.733, 0.758)	0.473	1.605	$\hat{\beta}_{28}$	(−0.162, 0.140, 0.319)	0.099	1.104
$\hat{\beta}_{14}$	(−0.456, −0.301, 0.100)	−0.219	0.803	$\hat{\beta}_{29}$	(−0.157, −0.066, 0.009)	−0.071	0.931

Table 6

Confusion matrix for online implementation.

	Predicted negative	Predicted positive
Actual negative	TN = 4972	FP = 17
Actual positive	FN = 1	TP = 10

Table 7

Performance comparison of online implementation with fuzzy logistic regression framework and crisp ML methods.

Method	Specificity	Sensitivity	F1 Score	MCC
Fuzzy logistic regression - Online stage	0.995	0.909	0.526	0.809
Crisp ML logistic regression	0.999	0.048	0.087	0.154
Crisp KNN ML method	0.999	0.009	0.019	0.071
Naive Bayes	0.999	0.097	0.171	0.258
Crisp ANN ML method	0.999	0.500	0.645	0.674
Crisp XG Boost ML method	0.999	0.500	0.645	0.674

a lower false positive (12 instances) but a much higher false negative (37 instances) decision rate.

Strelcenia and Pragoonwit (2023) discuss the increasing risk of fraud associated with credit card usage on online websites and platforms, proposing a knowledge-constrained generative adversarial network (K-CGAN) algorithm to detect fraud anomalies in transaction data. The K-CGAN model is evaluated using several machine learning algorithms; however, we benchmark against the XG-Boost method, which achieves the strongest classification results. At first glance, this method showcases high performance with an accuracy score of 99.9% and an F1 score of 0.999. All other machine learning models considered under the K-CGAN algorithm showcase similar results. However, there is no consideration of sensitivity and specificity to assess how overfitting caused by complete separation could create such artificially high performance (Charizanos et al., 2023). Additionally, the potential application of this system for real-time fraud detection and the level of accuracy in an online implementation are not considered.

The intelligent sampling approach of Chen et al. (2024) showcases strong F1 performance across all the long short-term memory (LSTM)-Conditional Random Field and three feed-forward deep neural network (FDNN) methods applied to the fraud detection dataset, which range between 0.800–0.830, compared to 0.526 for the fuzzy logistic regression framework. However, the sensitivity scores, here given as recall, indicate a significantly weaker classification performance. These range between 0.710–0.760, compared to the much stronger outcome from the fuzzy logistic regression with 0.909. This could be due to the imbalance and separation issues in the data; however, due to the lack of other performance measures provided by the authors, it is difficult to make a full assessment.

4. Application with simulated transaction data

4.1. Results of tuning and learning stages

Parameter tuning is performed using the same settings in Section 3.2. Table 2 of the Supplementary Material shows the tuning and learning outcomes in terms of MSE, MAE, and RMSE. We see a significant spread between the lowest and highest values achieved for these optimization measures. However, we see consistently low scores for all three optimization measures when analyzing the mean and standard deviation values using CV. This showcases consistently strong tuning outcomes with a small presence of outliers, which is evidenced further in Fig. 4 of the Supplementary Material. All three box plots indicate that a few scenarios produce outlier results for the optimization measures; however, the overall distribution of results clusters closely to the mean position. The selected optimal scenario has the following settings. Fuzzification factor m is set at 1.5. The asymmetry settings are taken as $l = 1$ and $r = 1$. The fuzzy classification threshold is $\bar{\tau} = (0.4, 0.5, 0.6)$. The fuzzification limits are 0 and 1.5. The coefficient generation intervals for β_0 and β_i are set to $[-2, 2]$.

Table 3 of the Supplementary Material presents the classification performance outcomes for the optimal solution extracted from the tuning and learning stage. We see similarly strong classification performance with all three optimization measures, with specificity at 0.936 indicating accurate classification towards the negative class and sensitivity of 0.992 indicating accurate classification towards the positive class. The lack of significant variability between specificity and sensitivity strongly indicates that the imbalance problem does not impact the model. Moreover, both F1 and MCC scores consistently indicate high classification performance despite separation and imbalance issues in the data.

Table 4 of the Supplementary Material shows TFN and crisp coefficient estimates obtained by the center of gravity method. There are two more pieces of evidence that point towards the proposed methodology's strong classification outcomes. Firstly, there are no large values for any of the coefficient estimates. Models impacted by complete separation often produce extremely large coefficient estimates for the predictors that cluster the binary responses. Secondly, none of the odds ratio estimates (exponential values for the coefficient estimates) deviates to infinity, which is a strong indicator of models not impacted by separation or imbalance.

4.2. Validation, online implementation and benchmarking

We implement the model using the $V = 5000$ most recent transactions in the selected dataset to examine the classification performance of the proposed online method using fuzzy logistic regression. The model continues to showcase strong efficiency, with an average time of 0.13 s to classify each transaction. Five benchmark ML methods on

Table 8
Online fuzzy logistic regression classification performance against benchmark studies.

Method	Specificity	Sensitivity	F1 Score	MCC	Accuracy
Online fuzzy logistic regression	0.995	0.909	0.526	0.809	99.6%
Fuzzy clustering - ANN (Behera & Panigrahi, 2015)					93.9%
Distributed tree-based (Vorobyev & Krivitskaya, 2022)	0.999	0.760	0.527	0.830	83.3%
K-CGAN method with XG-Boost (Strelcenia & Prakoonwit, 2023)			0.999		99.9%
Intelligent sampling (LSTM-CRF) (Chen et al., 2024)		0.760	0.800		
Intelligent sampling (FDNN, FR = 1.3) (Chen et al., 2024)		0.710	0.810		
Intelligent sampling (FDNN, FR = 1.4) (Chen et al., 2024)		0.770	0.810		
Intelligent sampling (FDNN, FR = 1.5) (Chen et al., 2024)		0.750	0.830		

Table 7 are also used for comparison for the simulated transactions data.

Table 5 of the Supplementary Material displays the performance measures for the fuzzy logistic regression method against the benchmark ML algorithms. All benchmark methods fail to produce valid results, with significant variability between sensitivity and specificity scores, which is evidence of the impact of imbalance problems. F1 scores and MCC remain low for all benchmark methods. However, the proposed online fuzzy logistic regression method achieves high performance outcomes with a specificity of 0.812, a sensitivity of 0.955, an F1 score of 0.811, and an MCC of 0.997. These strong, consistent results indicate no impact from imbalance or separation on the performance of the proposed classifier.

5. Conclusion and future work

In this study, we propose an innovative fraud detection method for better precision by focusing on the issues of handling non-stationary changes in fraud behavior, imbalance, and complete separation. To handle these issues, we implement a fuzzy logistic regression approach with tuning, learning, and online implementation stages.

Transaction fraud characteristics change rapidly, limiting predictive effectiveness given the non-stationarity of the mechanisms creating predictors and responses. The performance of the proposed solution shows that it is not affected by sample size and number of Monte Carlo replications. This is a key contribution and crucial benefit towards efficiency, providing a control factor towards non-stationary changes in the online implementation. This novelty also allows researchers to work with smaller sample sizes, enhancing the capacity for experimentation. In comparison, most methodologies in the literature focus on an implementation using the whole dataset, which can be inefficient and hence not applicable for real-time detection (Maes et al., 2002; Shen et al., 2007; Srivastava et al., 2008).

The results from performance analysis show that the proposed framework exhibits a high degree of accuracy in detecting both fraudulent and non-fraudulent transactions. There is no indication of negative influence on performance due to the existence of imbalance and complete separation in the fraud data, with consistently strong performance results on both tuning, learning, and online implementation stages. The model correctly classifies 90.9% of fraudulent (positive class) and 99.5% of non-fraudulent (negative class) transactions. We also see a low rate of false positive (transactions incorrectly classified as fraudulent) and false negative (transactions incorrectly classified as non-fraudulent) decisions. On the other hand, several approaches in the literature focus only on the predictive accuracy of positive classes (Behera & Panigrahi, 2015; Shen et al., 2007). It is crucial to utilize measures that assess both positive and negative classes, given that

a complete assessment of the impact of class imbalance can only be achieved by reviewing both classes of the binary response variable.

Compared to the recent fraud detection approaches, the proposed online fraud detection framework using fuzzy logistic regression outperforms the fuzzy clustering algorithm of Behera and Panigrahi (2015) on the predictive accuracy of fraudulent activity. At the same time, a complete assessment shows substantial accuracy for non-fraudulent transactions. While the results against the tree-based ML method of Vorobyev and Krivitskaya (2022) are fairly comparable in terms of granular false and true positive or negative rates, overall, the proposed framework has better sensitivity performance and imbalance handling. Moreover, Strelcenia and Prakoonwit (2023) shows strong accuracy and F1 scores, which could be impacted by the complete separation in the data. The separation problem would create over-fitting issues, making the performance scores almost perfect. On the other hand, the proposed fraud detection framework showcases strong performance towards both imbalance and separation issues.

The practical implication of the proposed framework is three-fold. Firstly, since its performance is not heavily dependent on the training sample size and all the software codes are readily available, it can easily be integrated into existing fraud monitoring systems in real-time. Thus, financial institutions can implement the proposed framework on their systems at a low cost. Secondly, the proposed framework captures the changing characteristics of both the fraudster and consumer sides. This provides the institutions with a sustainable system that needs periodic re-training at a frequency specified by the institution. Thirdly, our framework is highly precise in detecting fraudulent and non-fraudulent transactions. A higher performance in fraud detection with our framework implies less financial losses for a financial institution when implemented. It also means less account exposure to fraudsters when the financial institution fails to flag and stop the activity. Better precision in not interfering with legitimate transactions with our framework translates to higher consumer satisfaction and less consumer interaction, leading to a better operational environment for the institutions.

It should be noted that using unequally spaced time series data due to random sampling can introduce bias if not carefully handled. However, this is not an issue in our study, as this is handled by ensuring that the ranking of the data, or the order of transactions, is maintained and autocorrelation information is not utilized in the method.

A limitation of fraud detection studies is the unavailability of real and non-anonymized transaction datasets due to regulations on publishing financial data. Therefore, most studies either use the publicly available transaction fraud dataset or do not share their data.

Further research could focus on the methodology for cross-industry application of fuzzy classifiers for fraud prediction across different types of fraud. While we have mainly focused on transaction fraud,

other examples are identity theft, financial sanctions, and corporate fraud. Moreover, different types of fuzzy numbers could yield varied performance results. Experimentation with trapezoidal, Gaussian, L-R, and Z numbers could prove beneficial. Although our framework is mainly designed to identify fraudulent transactions, the odds ratios produced by the model at the end of the learning stage are useful to identify the most influential factors of suspicious transactions. The method applied in this study can be expanded with fuzzy hypothesis tests in future studies. Whether fuzzy regression parameters are significant or not can be investigated with fuzzy hypothesis tests, and the results can be used to identify the factors that cause doubt in a real-world banking application. However, this needs an unmasked dataset with information on the predictors.

CRedit authorship contribution statement

Georgios Charizanos: Conceptualization, Methodology, Software, Visualization, Investigation, Writing – original draft. **Haydar Demirhan:** Conceptualization, Methodology, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Duygu İçen:** Conceptualization, Methodology, Investigation, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All research data and codes are available via a GitHub repository.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.eswa.2024.124127>.

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